

Stat 3113: Lesson 4

Distributions of Quantitative Variables

Describing Data with Numerical Measures

Graphical methods may not always be sufficient for describing data.



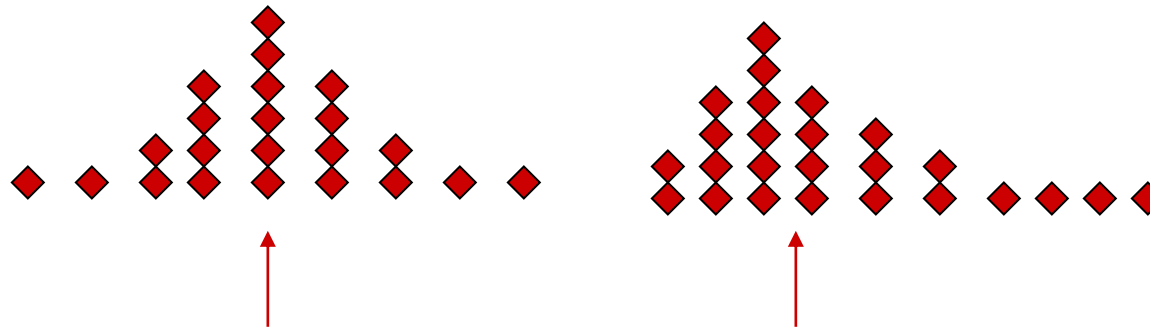
Numerical measures can be created for both populations and samples.

A **parameter** is a numerical descriptive measure calculated for a **population**.

A **statistic** is a numerical descriptive measure calculated for a **sample**.

Measures of Center

- A measure along the horizontal axis of the data distribution that locates the **center** of the distribution.



Dot plots

Arithmetic Mean or Average

- The **mean** of a set of measurements is the sum of the measurements divided by the total number of measurements.

$$\bar{x} = \frac{\sum x_i}{n}$$

where n = number of measurements

$\sum x_i$ = sum of all the measurements

- NB: The population mean is commonly denoted by μ .

Median

- The **median** of a set of measurements is the middle measurement when the measurements are ranked from smallest to largest.
- Notation: Sample median $\tilde{x}$ or M, population median $\tilde{\mu}$.
- The median is the 50th percentile – half of the data smaller and half larger.

Median

- *Quantile* → as decimal values.
- A **percentile** indicates where a value stands in comparison to others in a dataset. → as percent values.
- The 0.95 quantile point is exactly the same as the 95th percentile point.
- For instance, the 75th percentile means the value is higher than or equal to 75% of the data. It's useful for understanding rankings and distributions.
- For example, if exam scores are 70, 75, 80, 85, and 90, the 75th percentile score is 85, exceeding 75% of scores.

Rules: $n \times p = k$ → $x^{(k)}$ if integer
may be rounded up to next highest integer. → $\frac{x^{(k)} + x^{(k+1)}}{2}$ if not

$\frac{p \times (n+1)}{100}$ → If Integer → sample value in that position. → *Textbook definition*
→ If not → Average of sample values on either side.

Example

The number of quarts of milk purchased by 25 households:

0 0 1 1 1 1 1 2 2 2 2 2 2 2 2 3 3 3 3 3 4 4 4 5

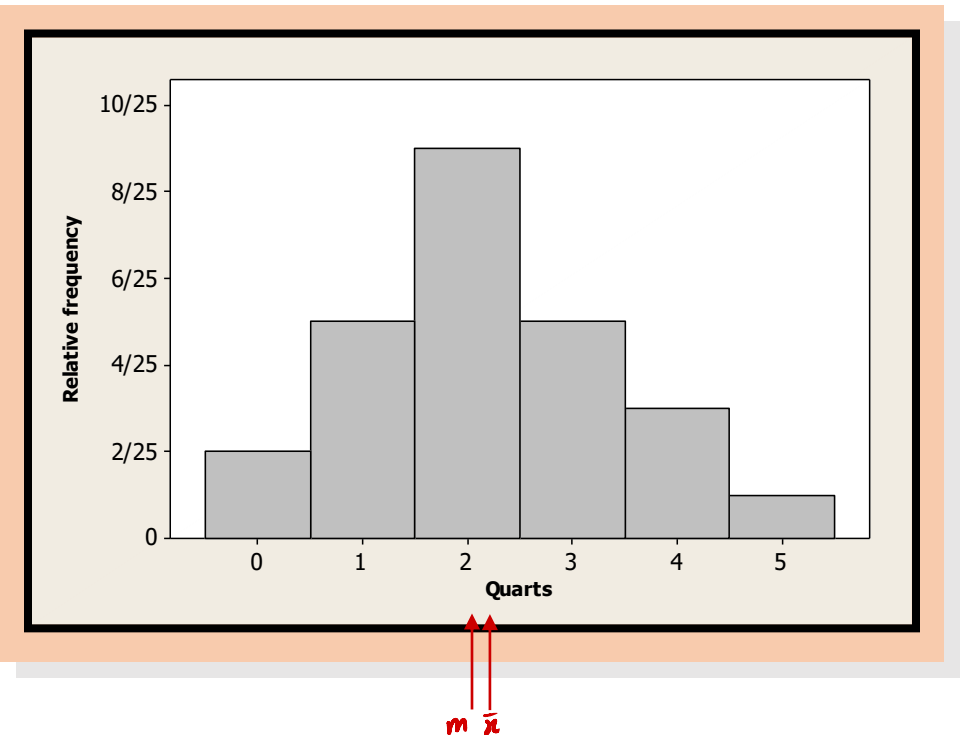
Histogram

- Mean?

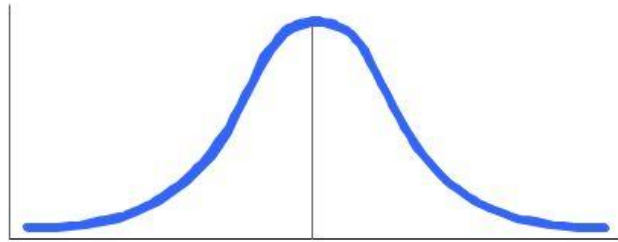
$$\bar{x} = \frac{\sum x_i}{n} = \frac{55}{25} = 2.2$$

- Median?

$$m = 2$$

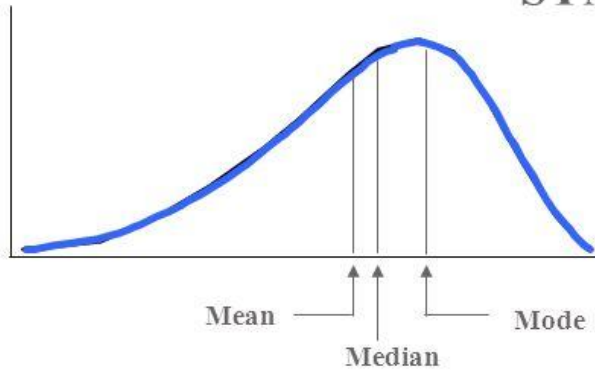


Skewness



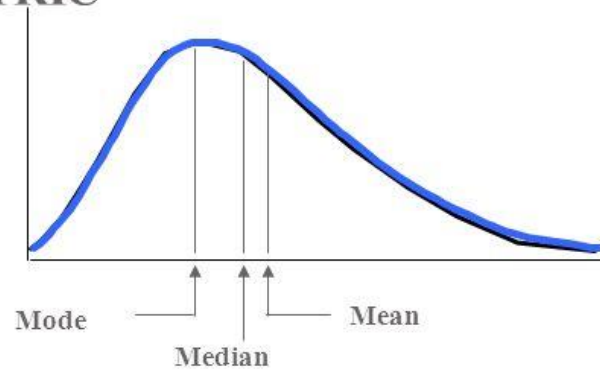
Mode = Mean = Median

SYMMETRIC



SKEWED LEFT

(negatively)



SKEWED RIGHT

(positively)

36

Measures
of Center and
Symmetry

Extreme Values

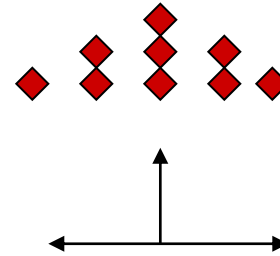
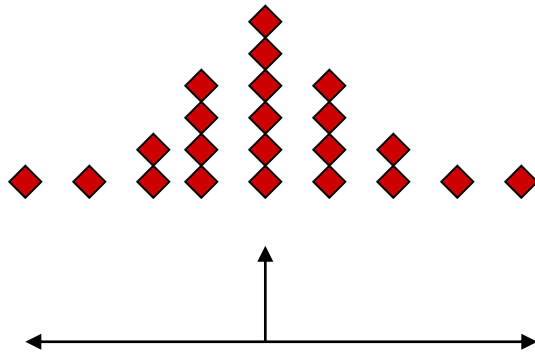
- The **median** is often used as a measure of center when the distribution is heavily skewed or has outliers.
- The mean is affected by extremely large or small values (outliers).
- The median is **NOT**, it is said to be resistant to outliers and skew.

Robust



Measures of Variability

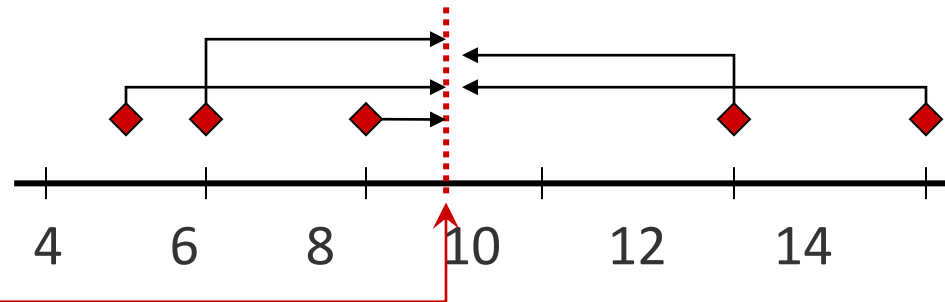
- A measure along the horizontal axis of the data distribution that describes the **spread/variation** of the distribution from the center.



The Variance

- The **variance** is a measure of variability that uses all the measurements. It measures the **average deviation of the measurements about their mean**.
- **Flower petals:** **5, 12, 6, 8, 14**

$$\bar{x} = \frac{45}{5} = 9$$



The Variance

- The **variance of a population** of N measurements is the average of the squared deviations of the measurements about their mean μ .

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

- The **variance of a sample** of n measurements is the sum of the squared deviations of the measurements about their mean, divided by $(n - 1)$.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

The Standard Deviation

- In calculating the variance, we squared all of the deviations, and in doing so changed the scale of the measurements.
- To return this measure of variability to the original units of measure, we calculate the standard deviation, the positive square root of the variance.

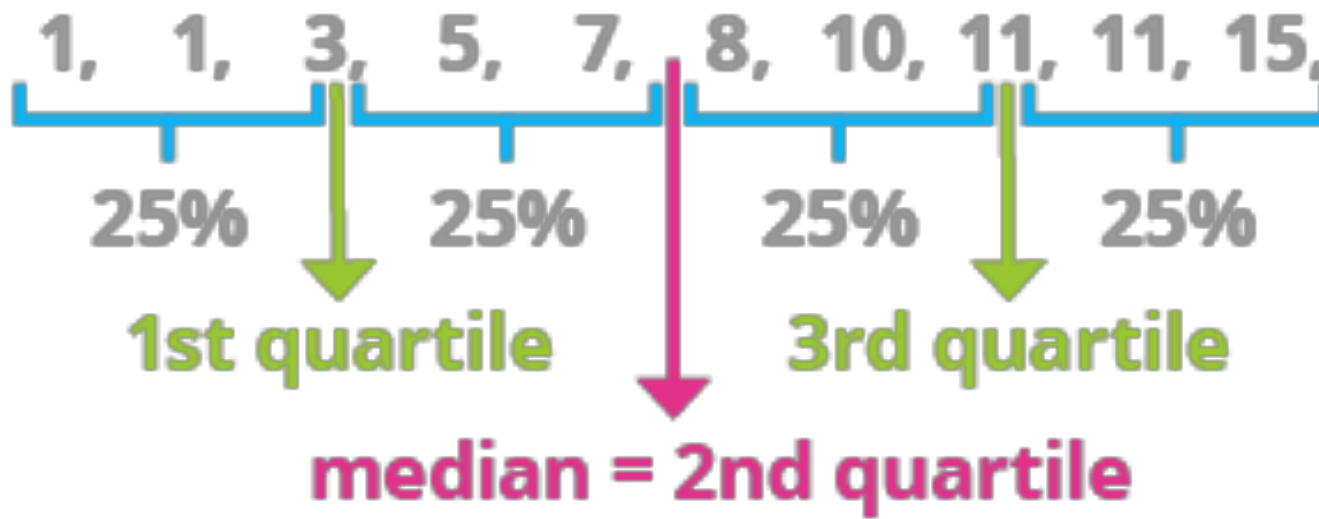
Population standard deviation : $\sigma = \sqrt{\sigma^2}$

Sample standard deviation : $s = \sqrt{s^2}$



Some Notes

- The value of s is **0** when all of the sample data is the same value.
- Otherwise, the value of s is positive.
- The larger the value of s^2 or s , the larger the variability of the data set.
- **Why divide by $n - 1$?**
 - The sample standard deviation s is often used to estimate the population standard deviation σ . Dividing by $n - 1$ gives us a better estimate of σ .



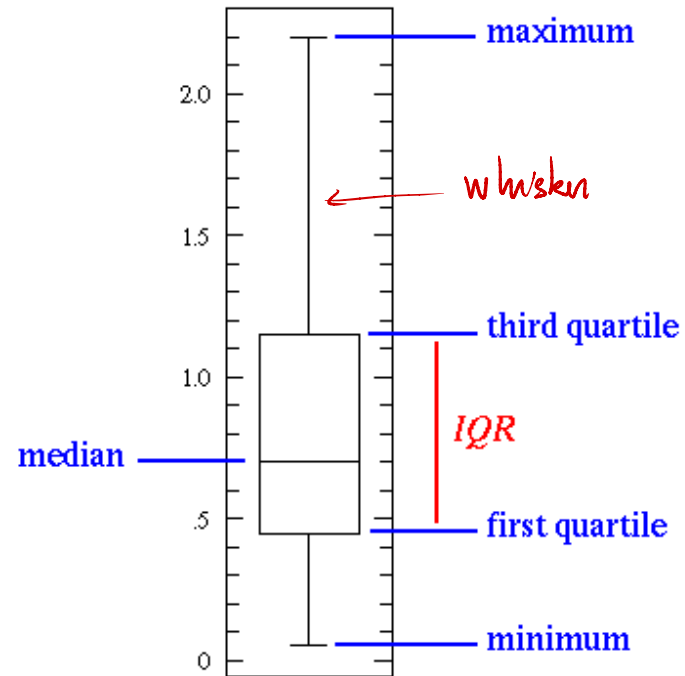
Divide the ordered data into quarters.
||

Quartiles and the IQR

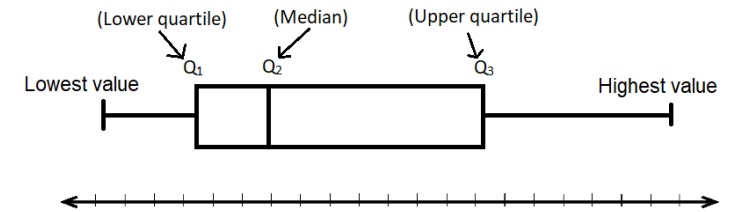
- The **first quartile (Q_1)** is the value of x which is larger than 25% and less than 75% of the ordered measurements.
- The **third quartile (Q_3)** is the value of x which is larger than 75% and less than 25% of the ordered measurements.
- The range of the “middle 50%” of the measurements is the **interquartile range, $IQR = Q_3 - Q_1$.**

Quantile Boxplots

Vertical layout



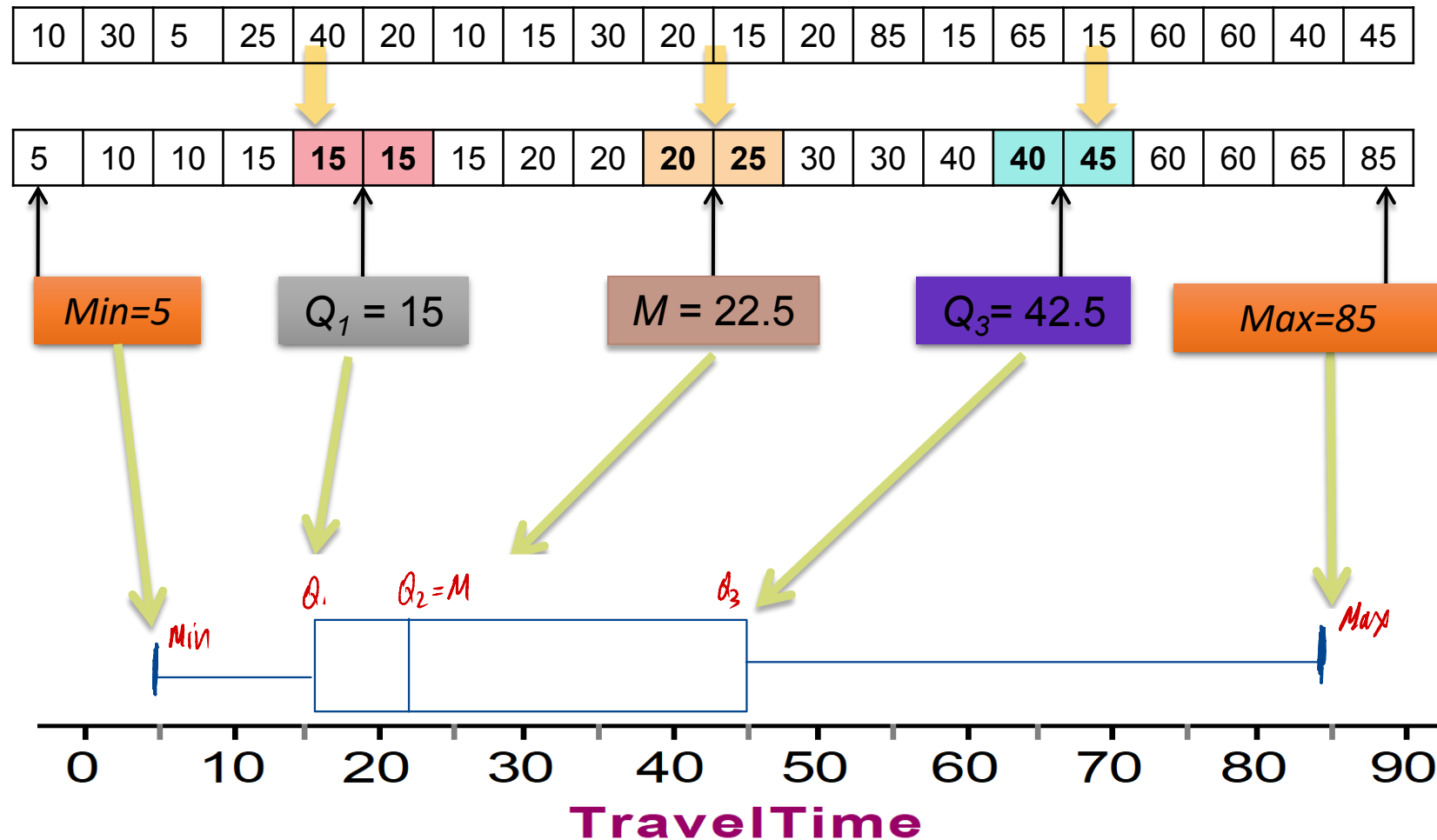
Horizontal layout



Five-number summary and boxplots

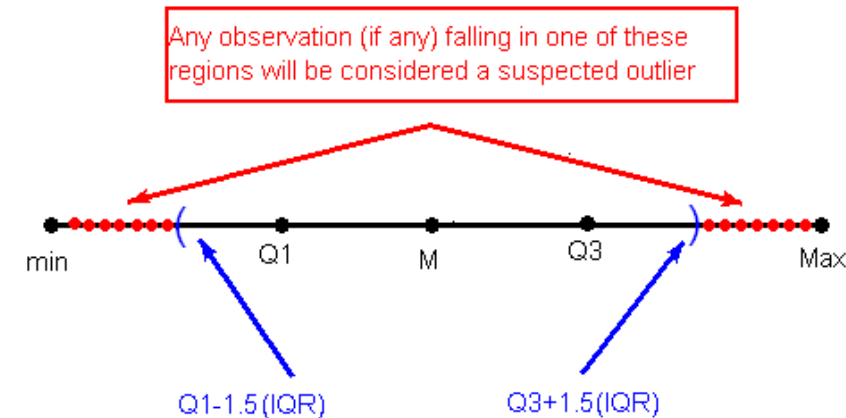
Consider a second travel times data set, these from New York.

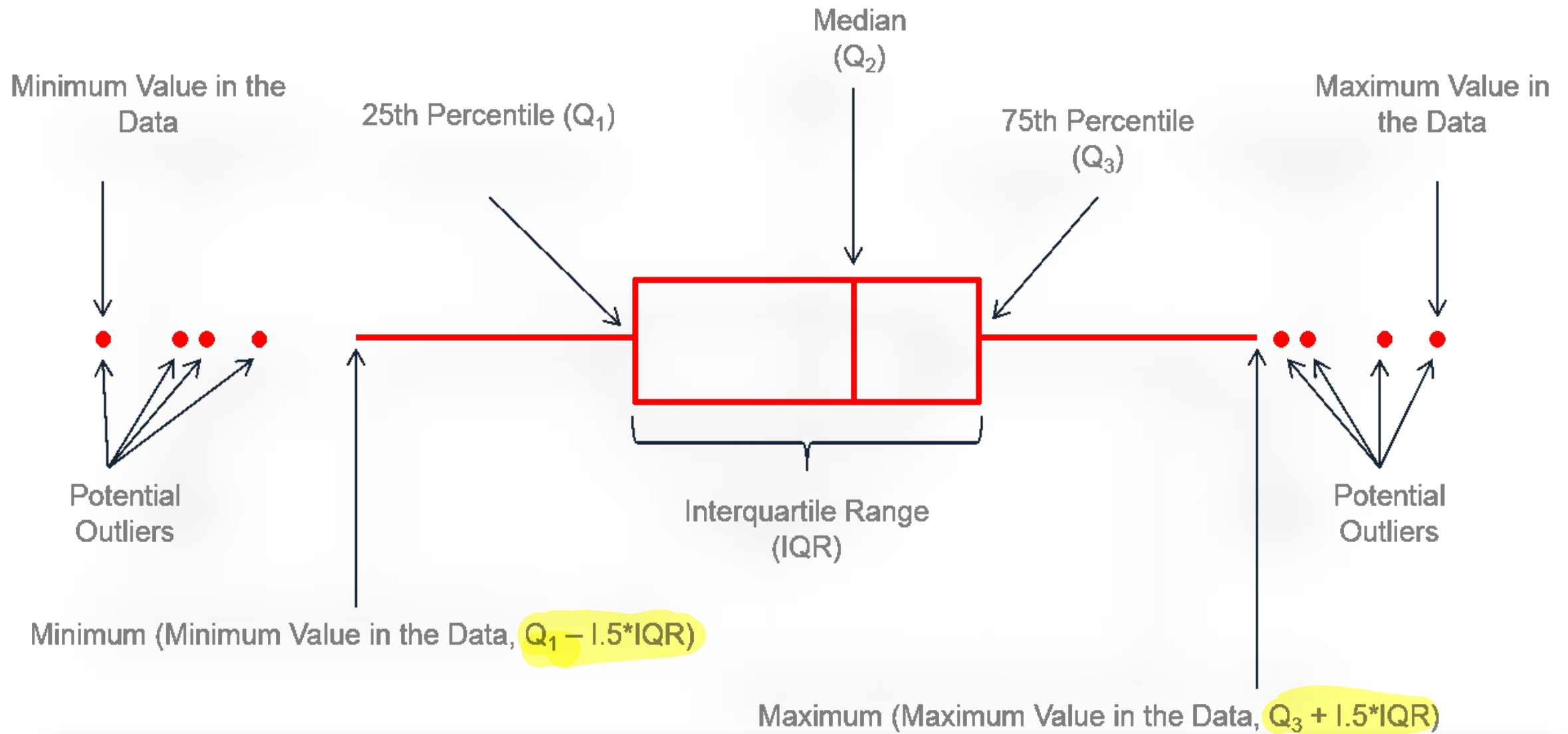
Find the five-number summary and construct a boxplot.



Spotting suspected outliers and modified boxplots

- $IQR = Q_3 - Q_1$
- In addition to serving as a measure of spread, the interquartile range (IQR) is used as part of a **rule of thumb** for identifying outliers.
- **The $1.5 \times IQR$ Rule for Outliers**
 - Call an observation a **suspected outlier** if it falls more than $1.5 \times IQR$ above the third quartile or below the first quartile.

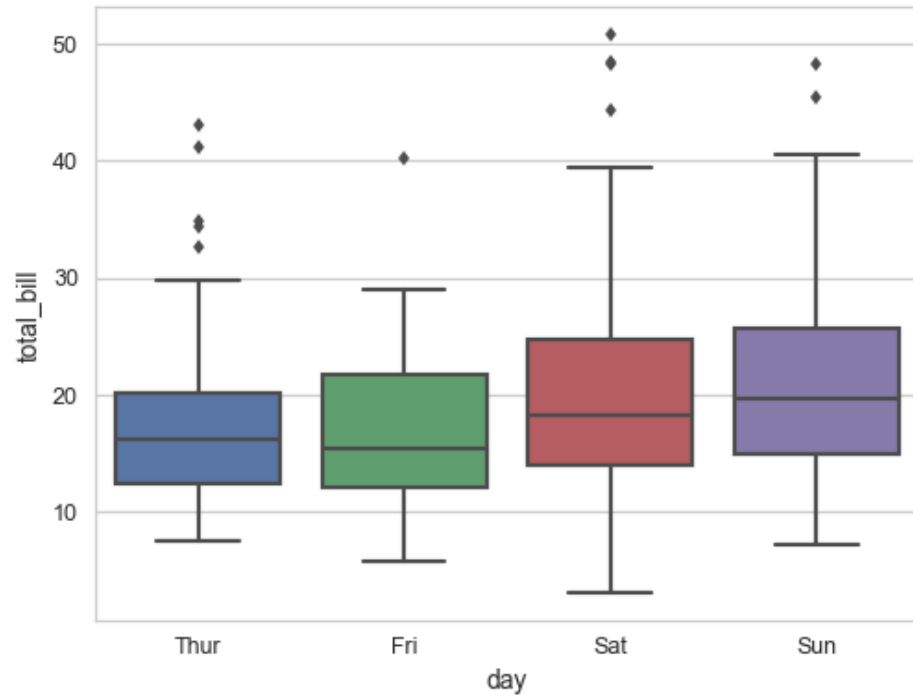




Boxplot with outliers

Parallel/ Side-by-Side Boxplots

Similarities / Dissimilarities!

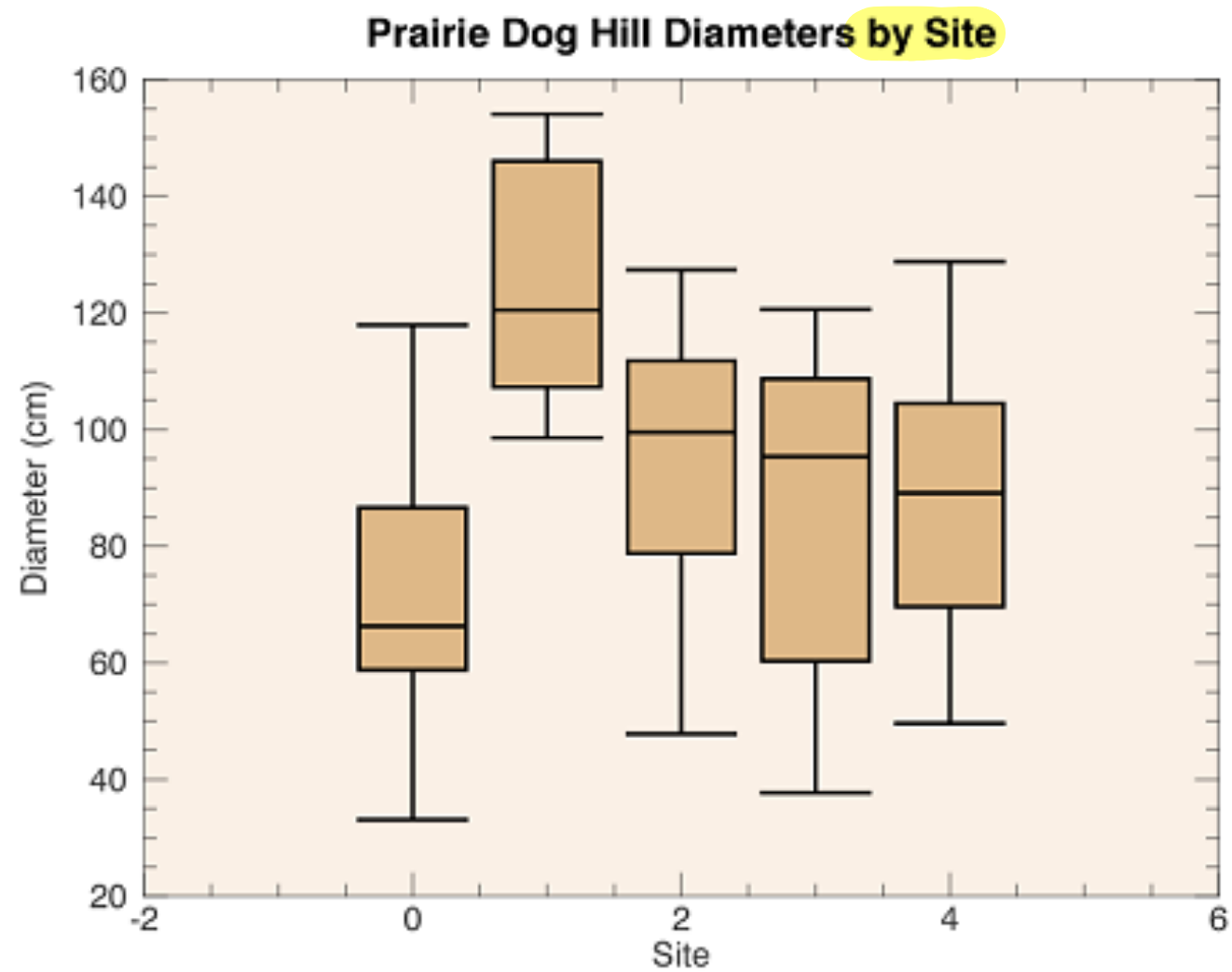


Great for comparing ^{ing} quantitative data across multiple groups.

** Guidelines for comparing boxplots*

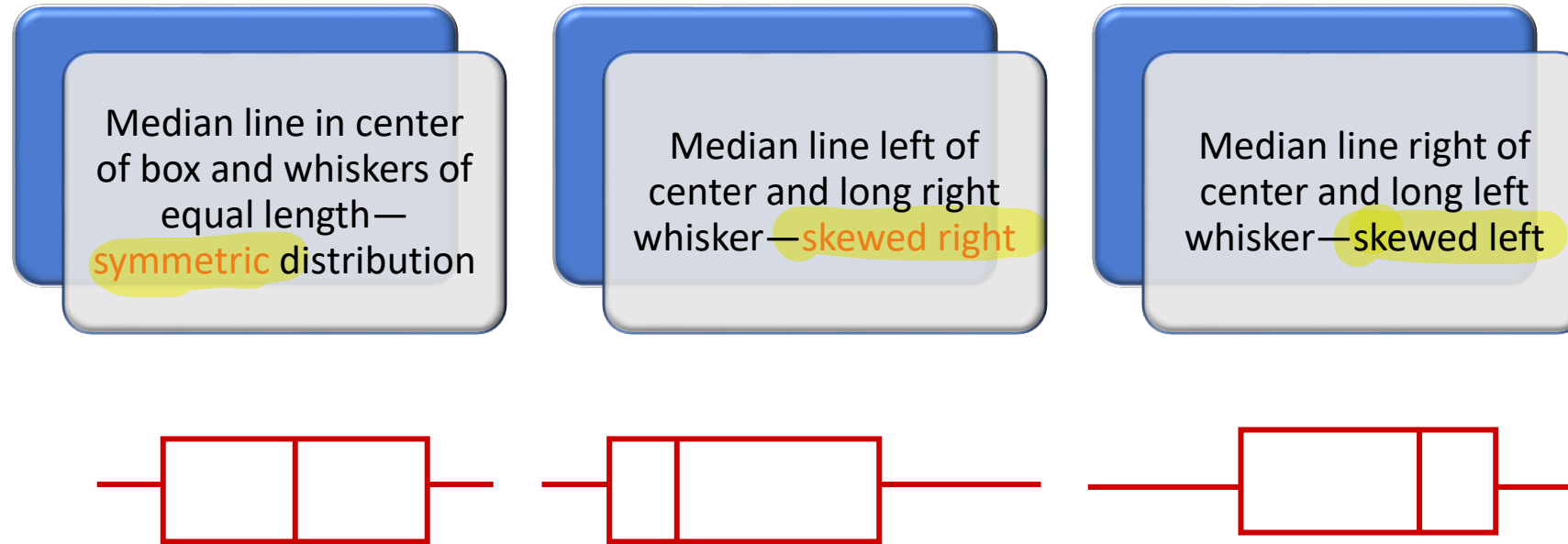
- 1) Compare the respective medians, to compare location*
- 2) Compare the interquartile ranges (box heights), to compare dispersion*
- 3) Look at the overall spread as shown by the adjacent values*
- 4) Look for signs of skewness.*
- 5) Look for potential outliers.*

→ Example in Lesson 5



Simulated Data

Interpreting Box Plots



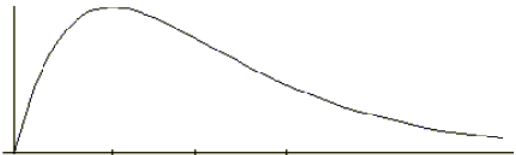
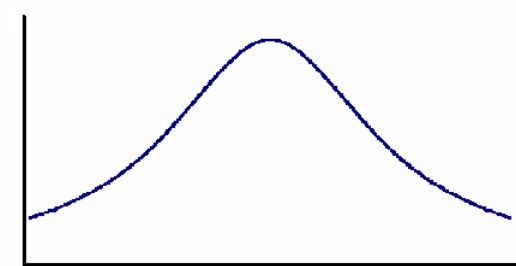
Note:

Boxplots are best used WITH a histogram or another graph like that!!!

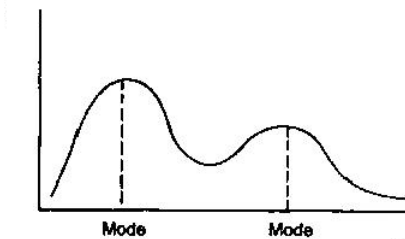
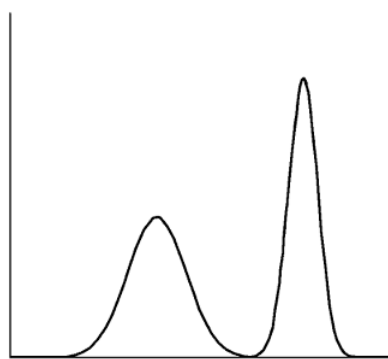
Five Aspects of the Overall Pattern of the Distribution of a Quantitative Variable

- **Outliers** – list points you feel are outliers or say “None” if there are no outliers.
- **Center** – central tendency of data: For the center, if there are no significant outliers or skew, you give *both the mean and median*.
- If there are outliers, give both the mean and median, **but mention you wouldn't use the mean due to outlier or skew!**

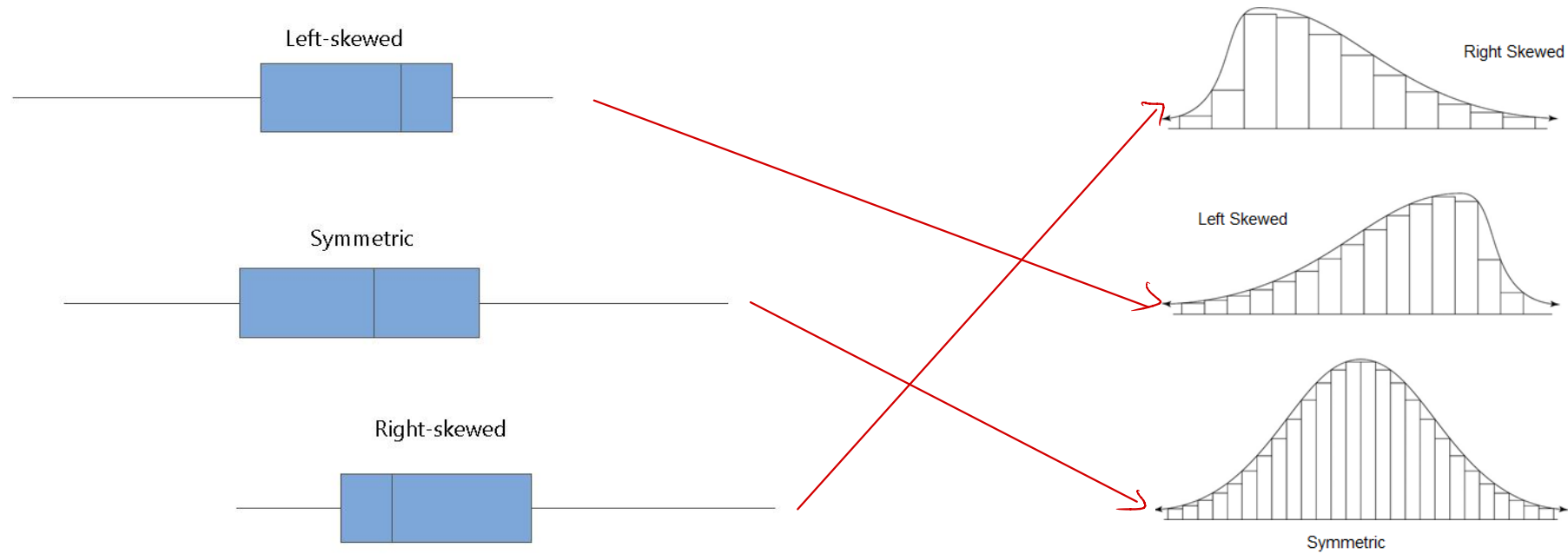
UNIMODAL



BIMODAL



- Spread – measure of variability: For the spread, give $Q1$ and $Q3$ and give the value of s .
- If there are outliers or skew, mention you wouldn't use the std dev due to the outlier!
- # groups –
 - ✓ Unimodal
 - ✓ Bimodal
 - ✓ Trimodal



Symmetry – Symmetric, Right Skewed, Left Skewed

Reading

Chapter 1 – Sections 1.2, 1.3



Next Topic



Distributions of Quantitative Variables – Examples & JMP Sessions



Chapter 1 - Sections 1.2, 1.3